

1 RNN + Attention vs. Transformer Architectures

1.1 RNN + Attention (Seq2Seq)

The classical sequence-to-sequence architecture consists of:

- An **encoder RNN** (LSTM or GRU) that processes the input sequence sequentially:

$$h_t^{\text{enc}} = \text{RNN}(x_t, h_{t-1}^{\text{enc}}).$$

- A **decoder RNN** that autoregressively generates output tokens:

$$s_t = \text{RNN}(y_{t-1}, s_{t-1}).$$

- **Additive or dot-product attention** over encoder states:

$$e_{t,i} = f_{\text{att}}(s_{t-1}, h_i^{\text{enc}}), \quad \alpha_{t,i} = \text{softmax}_i(e_{t,i}),$$

$$c_t = \sum_i \alpha_{t,i} h_i^{\text{enc}}.$$

- The decoder output distribution is conditioned on both the RNN state and context:

$$p(y_t \mid y_{<t}, x) = \text{softmax}(W[s_t, c_t]).$$

Key properties.

- Sequential computation in both encoder and decoder.
- Attention is *cross-attention only*: decoder attends to encoder states.
- Long-range dependencies handled via recurrent states + attention.
- Parallelism limited: $O(n)$ sequential steps.

1.2 Transformer

The Transformer removes recurrence entirely and relies on:

- **Self-attention** in encoder and decoder layers:

$$\text{SelfAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V,$$

where $Q = K = V$ are token representations of the same layer.

- **Cross-attention** in the decoder, where

$$Q = H^{\text{dec}}, \quad K = V = H^{\text{enc}}.$$

- **Position-wise feed-forward networks** applied identically at each position.
- **Positional encodings** to inject order information.

Key properties.

- No recurrence: full parallelism across sequence length.
- Encoder uses stacked self-attention layers only.
- Decoder uses *masked* self-attention + cross-attention.
- Complexity dominated by attention: $O(n^2)$ per layer.

1.3 High-Level Comparison

	RNN + Attention	Transformer
Core operation	Recurrence	Self-attention
Parallelism	Low	High
Long-range modeling	Via hidden state + attention	Direct via self-attention
Attention type	Cross-attention only	Self-attn + cross-attn (decoder)
Order information	Implicit in recurrence	Positional encodings
Complexity	$O(n)$ sequential steps	$O(n^2)$ per layer

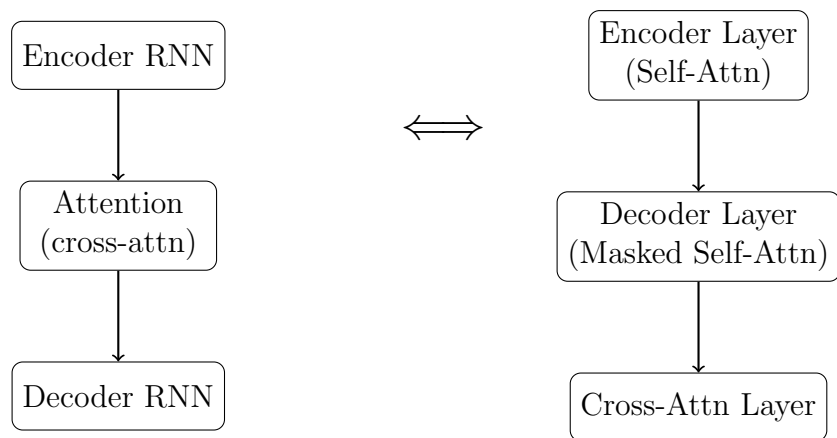


Figure 1: Architecture comparison between RNN + Attention (left) and Transformer (right).